Example 1.6

A manufacturer of ball-bearings, which are used in some machines, inspects to see whether the diameter of each ball-bearing is 5 mm. If the average diameter of ball-bearings is less than 4.75 mm or greater than 5.10 mm, then such ball-bearings will cause damages to the machine.

Here the null and the alternate hypotheses are

$$H_0 : \mu = 5 \text{ and } H_1 : \mu \neq 5.$$

Suppose that the decision on H_0 is made based on the diameter of 10 randomly selected ball-bearings. Let X_i, $i = 1, 2, \ldots, 10$ represent the diameter of the randomly chosen ball bearings. Then, the critical region is

$$\text{Critical Region} = \left\{ (x_1, x_2, \ldots, x_{10}) \mid \bar{x} = \frac{x_1 + x_2 + \ldots + x_{10}}{10} < 4.75 \text{ or } > 5.10 \right\}$$

In this case, $P\left(\bar{X} < 4.75\right)$ is the area, which will fall at the left end and $P\left(\bar{X} > 5.10\right)$ is the area, which will fall at the right end under the curve representing the sampling distribution of $\bar{X}$. This kind of test can be categorized as a two-tailed test (see Figure 1.3).

1.8 GENERAL PROCEDURE FOR TEST OF HYPOTHESES

The following steps constitute a general procedure, which can be followed for solving hypotheses testing problems based on both large and small samples.

Step 1 : Describe the population and its parameter(s). Frame the null hypothesis (H_0) and alternative hypothesis (H_1).

Step 2 : Describe the sample *i.e.*, data.

Step 3 : Specify the desired level of significance, α.

Step 4 : Specify the test statistic and its sampling distribution under H_0.

Step 5 : Calculate the value of the test statistic under H_0 for given sample.

Step 6 : Find the critical value(s) (table value(s)) from the statistical table generated from the sampling distribution of the test statistic under H_0 corresponding to α.

Step 7 : Decide on rejecting or not rejecting the null hypothesis based on the rejection rule which compares the calculated value(s) of the test statistic with the table value(s).

Now, let us see some of the large sample tests, which apply the above general procedure. As mentioned in Note-6, for large samples, the size of the sample is greater than or equal to 30. In the case of two samples considered for a hypotheses testing problem, the test is a large sample test, when the sizes of **both** the samples are greater than or equal to 30.

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